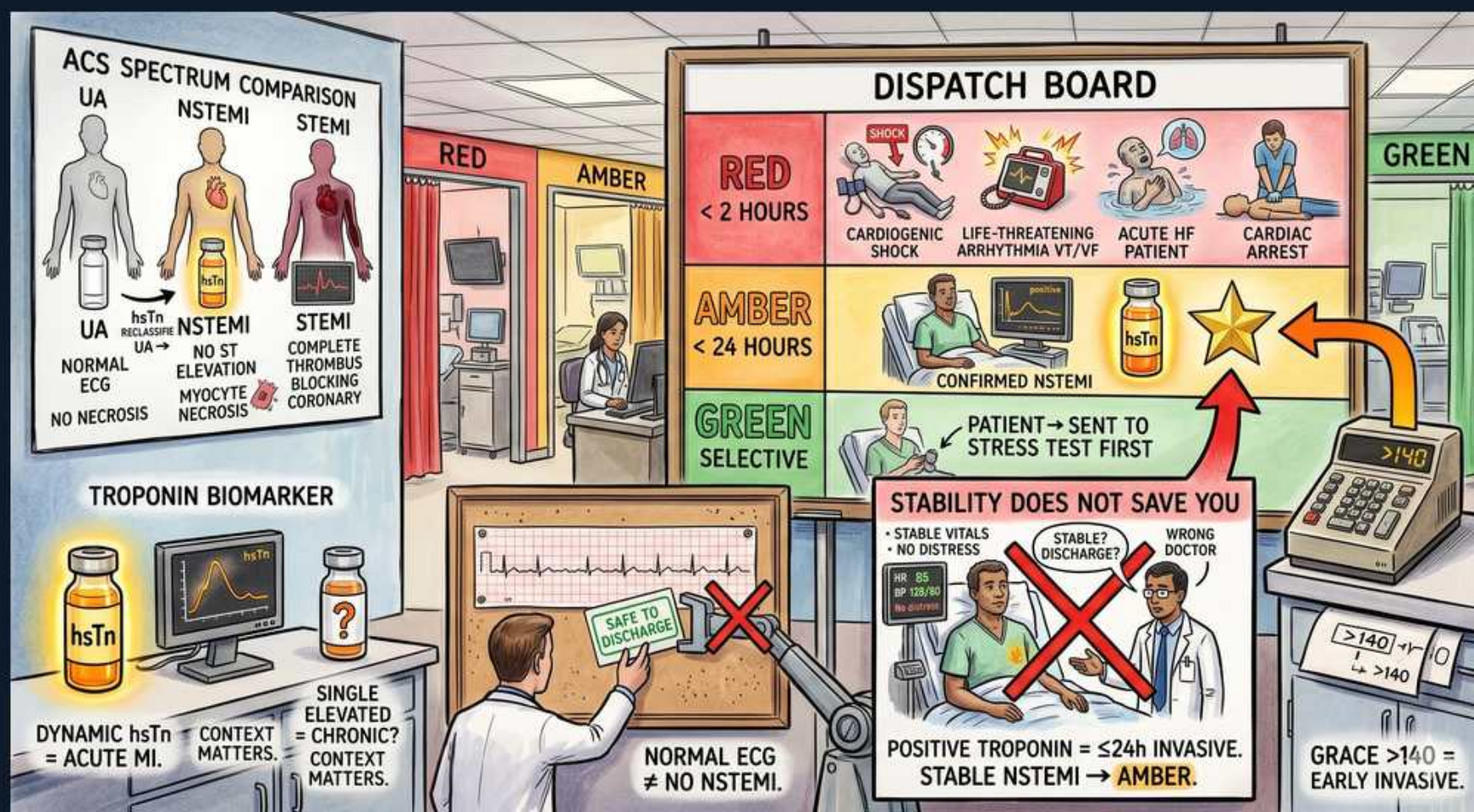


IHD / ACS — NSTEMI-ACS Pathophysiology & Spectrum



1. ACS spectrum

WHAT YOU SEE

The 'ACS SPECTRUM COMPARISON' poster on the left wall showing three body figures side by side — UA, NSTEMI, STEMI — lined up as one continuum, the leftmost grey/normal heart progressing to the rightmost fully-blocked red heart.

WHAT IT ENCODES

Acute coronary syndrome is a single pathophysiologic continuum of coronary atherothrombosis: an atherosclerotic plaque ruptures/erodes, exposing thrombogenic lipid core and tissue factor, triggering platelet adhesion and thrombus. The three faces differ by degree of occlusion and downstream necrosis — unstable angina (partial occlusion, NO necrosis), NSTEMI (partial occlusion WITH necrosis), STEMI (complete occlusion, transmural necrosis). UA and NSTEMI are grouped as NSTE-ACS because they share management; the discriminator between them is troponin.

BOARD TRAP

WRONG answer: treating UA, NSTEMI, and STEMI as three separate diseases with unrelated causes. Discriminator — all three stem from the same plaque-rupture/thrombus mechanism; they are points on a severity spectrum. This is why aspirin, P2Y12, anticoagulation, and statins apply across the board, while only complete-occlusion STEMI gets emergent reperfusion.

2. UA — no necrosis

WHAT YOU SEE

UA figure: NORMAL ECG, NO NECROSIS, hsTn negative

WHAT IT ENCODES

Unstable angina = ischemic symptoms (rest pain, crescendo angina, or new-onset severe angina) from a partially occluding/non-occlusive thrombus, but ischemia is brief/incomplete enough that there is NO myocyte necrosis → troponin remains NEGATIVE. The ECG may show ST depression or T-wave inversion or be normal. With high-sensitivity troponin assays, true UA is now diagnosed far less often because subtle necrosis is detected.

BOARD TRAP

WRONG answer: continuing to call it UA after a serial troponin comes back elevated. Discriminator — the instant troponin rises with a dynamic pattern, the diagnosis becomes NSTEMI by definition. UA is a troponin-negative diagnosis; necrosis (positive troponin) reclassifies it.

3. NSTEMI — necrosis, no ST elevation

WHAT YOU SEE

NSTEMI figure: NO ST elevation, MYOCYTE NECROSIS

WHAT IT ENCODES

NSTEMI = myocyte necrosis (troponin POSITIVE with a dynamic rise/fall) caused by a partial or transiently complete coronary thrombus, usually producing SUBENDOCARDIAL injury (the subendocardium is the most flow-dependent, last-perfused layer). There is NO persistent ST elevation; the ECG typically shows ST depression and/or T-wave inversion. Subendocardial (not transmural) injury is why ST elevation is absent.

BOARD TRAP

WRONG answer: using the ECG to separate NSTEMI from UA. Discriminator — both NSTE-ACS entities can have identical ECGs (ST depression, T inversion, or normal); it is the TROPONIN that distinguishes them — positive = NSTEMI, negative = UA. ECG separates STE (STEMI) from NSTE, not NSTEMI from UA.

4. STEMI — complete occlusion

WHAT YOU SEE

STEMI figure: complete thrombus blocking coronary

WHAT IT ENCODES

STEMI = sudden COMPLETE coronary occlusion by a fibrin/platelet thrombus → full-thickness TRANSMURAL ischemia → persistent ST-segment ELEVATION (or new LBBB) in contiguous leads, then troponin rise. It mandates IMMEDIATE reperfusion: primary PCI within 90 min of first medical contact (or 120 min if transfer), or fibrinolysis within 30 min if PCI is unavailable within that window.

BOARD TRAP

WRONG answer: giving fibrinolytics for an NSTEMI to 'open the artery faster.' Discriminator — fibrinolysis benefits ONLY the complete-occlusion STEMI; in NSTE-ACS thrombolytics increase bleeding/mortality without benefit (the artery is not fully occluded). The reperfusion-now rule belongs to STEMI alone.

5. hsTn dynamic = acute MI

WHAT YOU SEE

Troponin biomarker monitor; DYNAMIC hsTn → ACUTE MI

WHAT IT ENCODES

The defining biomarker feature of ACUTE MI is a DYNAMIC rise and/or fall of high-sensitivity cardiac troponin with at least one value above the 99th-percentile upper reference limit, in the right clinical context (symptoms, ECG, or imaging). Rapid 0/1h or 0/2h hs-troponin algorithms use absolute delta thresholds to rule in or rule out MI quickly. The 'change over time' is what signals acute injury.

BOARD TRAP

WRONG answer: diagnosing acute MI from one elevated troponin alone. Discriminator — a single high troponin without a dynamic delta may be chronic injury; conversely a normal first value does not exclude early MI. The acute MI diagnosis requires the dynamic pattern plus clinical context, not an isolated number.

6. Single elevated = chronic? context

WHAT YOU SEE

SINGLE ELEVATED = CHRONIC? CONTEXT MATTERS

WHAT IT ENCODES

A STATIC (non-changing) elevated troponin usually reflects CHRONIC myocardial injury rather than acute MI — common causes include CKD (reduced clearance), chronic HF, LV hypertrophy, and structural heart disease. Distinguish acute MI (type 1, plaque rupture) and type-2 MI (supply-demand mismatch) from non-ischemic troponin elevation (myocarditis, PE, sepsis, tachyarrhythmia) using the trajectory of serial values and the clinical picture.

BOARD TRAP

WRONG answer: diagnosing and treating NSTEMI for any elevated troponin in a CKD patient. Discriminator — baseline chronic elevation is common in renal failure; only a DYNAMIC change from that baseline plus ischemic context indicates acute MI. Treating a stable chronic elevation as ACS leads to unnecessary anticoagulation/cath and bleeding risk.

7. Normal ECG ≠ not NSTEMI

WHAT YOU SEE

ECG strip; NORMAL ECG ≠ NO NSTEMI

WHAT IT ENCODES

A normal or non-diagnostic initial ECG does NOT exclude NSTEMI — up to a third of NSTEMI patients have a non-diagnostic first tracing, and posterior or circumflex-territory infarcts are electrically silent on the standard 12 leads. Obtain SERIAL ECGs (and posterior leads V7–V9 if posterior MI suspected) plus serial high-sensitivity troponins before excluding ACS.

BOARD TRAP

WRONG answer: discharging a patient with classic ischemic chest pain because the first ECG is normal. Discriminator — the ECG can be normal early or with subtle/posterior territory involvement; ruling out ACS requires serial ECGs AND serial troponins over time, not a single normal tracing. This is a classic 'missed MI' malpractice scenario.

8. ECG: ST depression / T inversion

WHAT YOU SEE

The bedside ECG monitor beside the clinician showing the NSTEMI-ACS tracing — ST depression / T-wave inversion (no ST elevation), the electrical signature of subendocardial ischemia.

WHAT IT ENCODES

The characteristic NSTEMI-ACS ECG findings are horizontal/downsloping ST DEPRESSION (≥ 0.5 –1 mm) and/or symmetric T-WAVE INVERSION, reflecting subendocardial ischemia — but with NO persistent ST elevation. Deep symmetric precordial T-wave inversions (Wellens' pattern in V2–V3) signal a critical proximal LAD lesion and are high-risk. ST depression in many leads with aVR elevation suggests left-main or multivessel disease.

BOARD TRAP

WRONG answer: calling new persistent ST elevation an 'NSTEMI variant.' Discriminator — persistent ST elevation (or new LBBB) defines STEMI and triggers the emergent-reperfusion pathway; ST depression/T inversion define NSTEMI-ACS managed by risk-stratified invasive timing. Don't blur the two — the reperfusion urgency differs completely.

9. GRACE >140 → early invasive

WHAT YOU SEE

GRACE >140 → EARLY INVASIVE register tag

WHAT IT ENCODES

The GRACE score (age, HR, SBP, creatinine, Killip class, cardiac arrest at admission, ST deviation, elevated troponin) risk-stratifies NSTEMI-ACS for in-hospital and 6-month mortality. A GRACE >140 marks HIGH risk and warrants an EARLY invasive strategy (angiography within 24h). GRACE 109–140 = intermediate; <109 = low. TIMI and HEART scores are alternative risk tools (HEART for ED chest-pain disposition).

BOARD TRAP

WRONG answer: interpreting a low GRACE/TIMI score as 'no significant disease, safe to discharge.' Discriminator — these scores guide the TIMING/intensity of invasive workup and prognosis, not whether the patient has ACS. A troponin-positive NSTEMI still needs admission and antithrombotic therapy even if the score is modest.

10. Stability does not save you

WHAT YOU SEE

STABILITY DOES NOT SAVE YOU panel; stable vitals, no distress

WHAT IT ENCODES

Hemodynamic stability and feeling well do NOT downgrade a confirmed NSTEMI — a comfortable patient with normal vitals and a positive troponin still has acute myocardial necrosis from an active coronary culprit and remains at risk of reocclusion, arrhythmia, and death. They require admission, DAPT + anticoagulation + statin, and a risk-stratified invasive strategy (within 24h if high-risk).

BOARD TRAP

WRONG answer: discharging or merely observing a stable-appearing NSTEMI because they look fine. Discriminator — symptom resolution and stable vitals do not mean the culprit lesion is healed; troponin positivity mandates inpatient management. 'Stable NSTEMI' is still NSTEMI.

11. Positive troponin → ≤24h invasive

WHAT YOU SEE

POSITIVE TROPONIN = ≤24h INVASIVE; STABLE NSTEMI → AMBER

WHAT IT ENCODES

A high-risk NSTEMI (positive troponin with dynamic change, GRACE >140, or dynamic ST/T changes) should undergo coronary angiography within 24h (early invasive). A stable confirmed NSTEMI falls in the AMBER/<24h tier rather than the immediate <2h tier. Intermediate-risk patients may go within 24–72h; genuinely low-risk patients get ischemia-guided testing.

BOARD TRAP

WRONG answer: rushing every NSTEMI to primary PCI within 90 minutes like a STEMI. Discriminator — NSTEMI timing is measured in HOURS (typically <24h for high-risk), not minutes, because there is no complete occlusion to reopen emergently. The 90-minute door-to-balloon rule belongs to STEMI; applying it to NSTEMI is a category error.

12. RED <2h — immediate

WHAT YOU SEE

RED <2 HOURS row: cardiogenic shock, VT/VF, acute HF, cardiac arrest

WHAT IT ENCODES

Very-high-risk NSTEMI-ACS features mandate an IMMEDIATE (<2h) invasive strategy regardless of the risk score: hemodynamic instability/cardiogenic shock, recurrent or refractory ischemic chest pain despite medical therapy, life-threatening arrhythmia (sustained VT/VF) or cardiac arrest, acute heart failure, and mechanical complications (acute MR, VSD, free-wall rupture). These override GRACE/TIMI entirely.

BOARD TRAP

WRONG answer: waiting to calculate a GRACE score before activating the cath lab in a shocked or arresting NSTEMI-ACS patient. Discriminator — the very-high-risk clinical features bypass risk scoring and go straight to <2h angiography; risk calculators are for the stable patients deciding between <24h and ischemia-guided pathways.

13. GREEN — selective / stress first

WHAT YOU SEE

GREEN SELECTIVE row: patient sent to stress test first

WHAT IT ENCODES

Genuinely LOW-risk patients — no recurrent symptoms, NEGATIVE serial troponins, no dynamic ECG changes, low GRACE/TIMI — can take an ISCHEMIA-GUIDED (selective/conservative) strategy: non-invasive stress testing (exercise ECG, stress imaging, or coronary CT angiography) first, reserving invasive angiography for those with provoked ischemia or recurrent symptoms.

BOARD TRAP

WRONG answer: sending a troponin-positive NSTEMI to an outpatient stress test instead of angiography. Discriminator — ischemia-guided/selective testing is ONLY for the truly low-risk, ruled-out patient; a confirmed NSTEMI (positive troponin) requires an invasive strategy, and stress testing an actively ischemic patient is both dangerous and inappropriate.
